

Task 5: Transaction-Based Payment Simulation

Aim: To simulate an online payment system using database transactions by deducting the amount from a user account and adding it to a merchant account, ensuring data consistency using COMMIT and ROLLBACK operations.

Algorithm:

1. Start the program
2. Enter payment amount
3. Begin transaction
4. Check user account balance
5. If balance is sufficient
 - Deduct amount from user
 - Add amount to merchant
6. If all operations successful → COMMIT
7. If any error occurs → ROLLBACK
8. Display success or failure message
9. Stop the program

Program:

db.php:

```
<?php
$conn = new mysqli("localhost", "root", "", "bank_db");
if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}
?>
```

Payment.php:

```
<?php
include 'db.php';
if ($_SERVER["REQUEST_METHOD"] == "POST") {
    $amount = $_POST['amount'];

    // Start Transaction

    $conn->begin_transaction();
```

```

try {
    // Get User Balance
    $result = $conn->query("SELECT balance FROM accounts WHERE name='User'");
    $row = $result->fetch_assoc();
    $user_balance = $row['balance'];
    // Check Balance
    if ($user_balance < $amount) {
        throw new Exception("Insufficient Balance!");
    }
    // Deduct from User
    $conn->query("UPDATE accounts SET balance = balance - $amount WHERE
name='User'");
    // Add to Merchant
    $conn->query("UPDATE accounts SET balance = balance + $amount WHERE
name='Merchant'");
    // Commit Transaction
    $conn->commit();
    $message = "Payment Successful!";
} catch (Exception $e) {
    // Rollback Transaction
    $conn->rollback();
    $message = "Transaction Failed: " . $e->getMessage();
}
}

// Fetch Updated Balances
$result = $conn->query("SELECT * FROM accounts");
?>

<!DOCTYPE html>

<html>

<head>

```

```
<title>Payment System</title>

<style>

    body { font-family: Arial; background: #f2f2f2; }

    .box {

        width: 350px;

        margin: 100px auto;

        background: white;

        padding: 20px;

    }

    input, button {

        width: 100%;

        padding: 10px;

        margin: 10px 0;

    }

    .msg {

        font-weight: bold;

        color: blue;

    }

</style>

</head>

<body>

<div class="box">

    <h2>Payment Simulation</h2>

    <form method="POST">

        <input type="number" name="amount" placeholder="Enter Amount" required>

        <button type="submit">Pay</button>

    </form>

    <?php if(isset($message)) echo "<p class='msg'>$message</p>"; ?>

</div>

</body>

</html>
```

```
<h3>Account Balances</h3>

<?php while($row = $result->fetch_assoc()) { ?>
    <p><?php echo $row['name']; ?>: ₹<?php echo $row['balance']; ?></p>
<?php } ?>
</div>
</body>
</html>
```

Database setup:

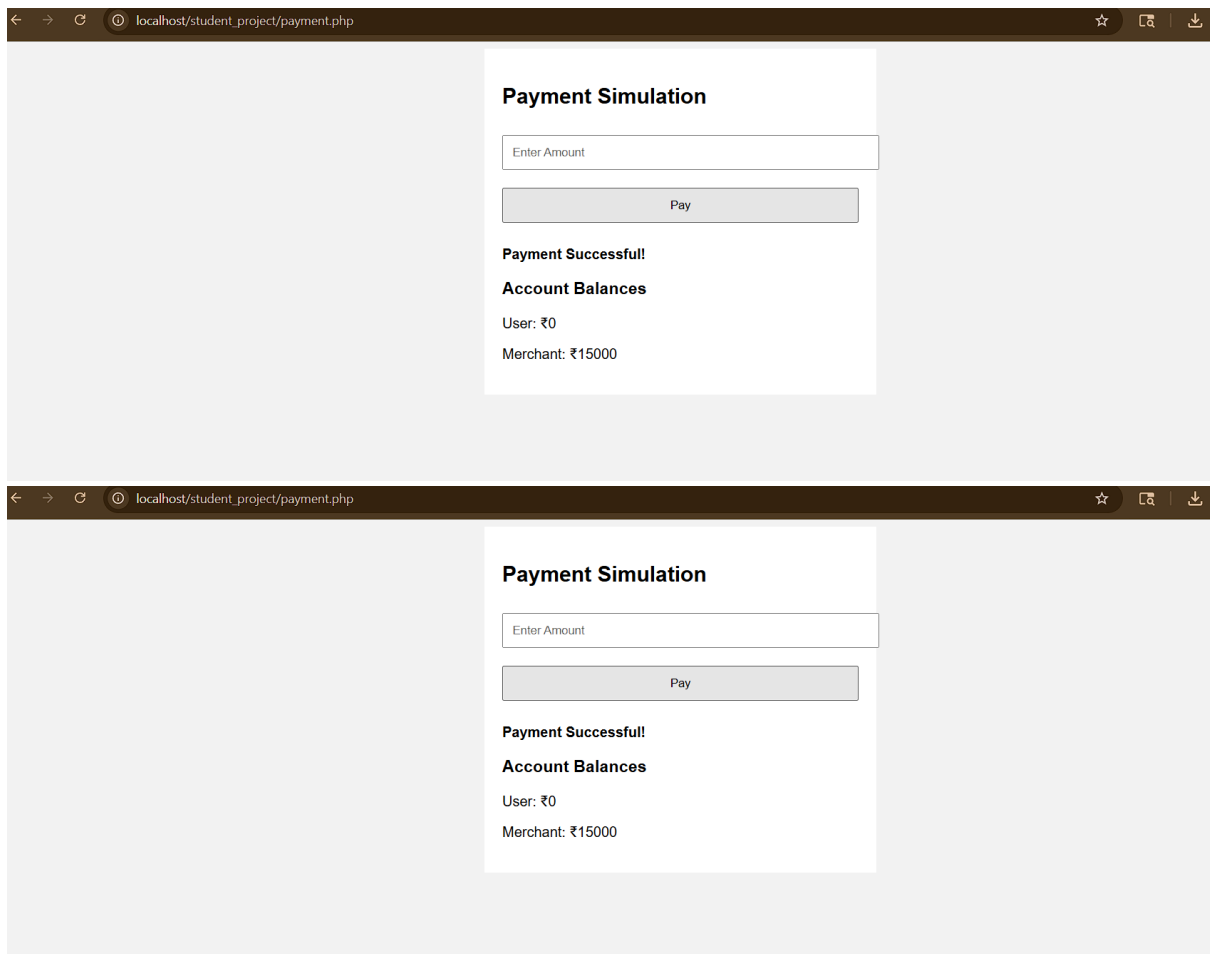
```
CREATE DATABASE bank_db;

USE bank_db;

CREATE TABLE accounts (
    id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(50),
    balance INT
);

INSERT INTO accounts (name, balance) VALUES
('User', 10000),
('Merchant', 5000);
```

Output:



The image displays two identical screenshots of a web browser window, stacked vertically. The browser's address bar shows the URL 'localhost/student_project/payment.php'. The page content is centered and features a white box with a light gray border. Inside this box, the title 'Payment Simulation' is at the top. Below it is a text input field with the placeholder 'Enter Amount'. Underneath the input field is a gray button labeled 'Pay'. Below the button, the text 'Payment Successful!' is displayed. This is followed by the heading 'Account Balances'. At the bottom of the box, the current balances are shown: 'User: ₹0' and 'Merchant: ₹15000'.

Payment Simulation

Enter Amount

Pay

Payment Successful!

Account Balances

User: ₹0

Merchant: ₹15000

Result: Thus, the execution of the transaction based payment simulation has been successfully executed and output generated .